Managing the risks of energy industry

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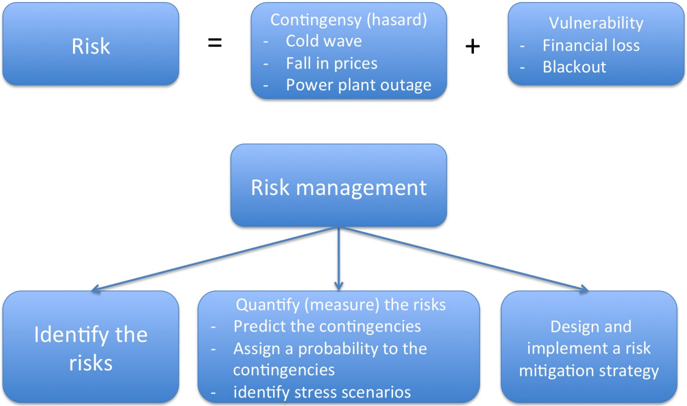
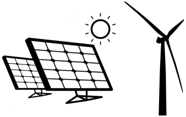
Overview of the course

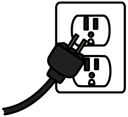
- The course focuses on risks and risk management for the electrical energy industry.



Risks of the electrical energy industry



Intermittent production



Electricity consumption

Distribution networks

Electricity markets

Overview of the course

1. Overview of features and risks of the electrical energy industry
2. Managing electricity market risks
3. Managing the risks of intermittent renewable production
4. Managing the risks of electricity consumption
5. Case studies in energy risk management

Evaluation

Evaluation consists of QCM and data analysis mini-project (list of projects was sent by email).

Submit deliverable (report + code or detailed notebook) on or before Jan 23.

Project reports will be evaluated based on

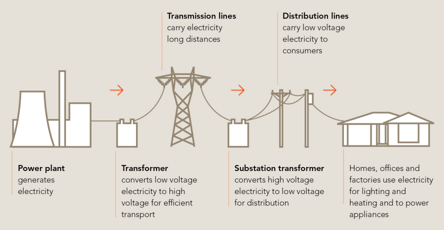
- Understanding of the energy risk management problem
- Clarity of model description
- Interpretation of the results
- Quality of visualizations

Follow AI use guidelines in Réglement de Scolarité (use allowed as writing or coding aid, but not for report generation, include AI use statement)

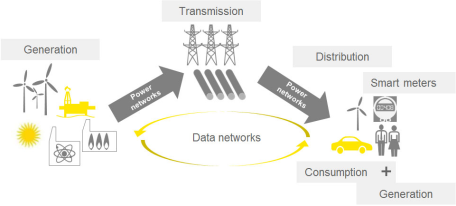
Some groups may be invited for oral presentaiton if I suspect AI overuse

Outline

The electricity supply chain



Source: `eex.gov.au`



Source: `ey.com`

Historically, energy production and distribution was carried out by integrated energy companies; electricity was produced by steam turbines at large power plants and sold to consumers at regulated tariffs

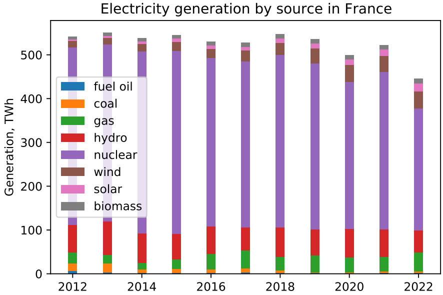
Nowadays, electricity production is competitive, but transport and distribution continue to be managed by state companies; distributed renewable generation plays an important role and data networks and smart meters are used for load management

The electricity setor in France

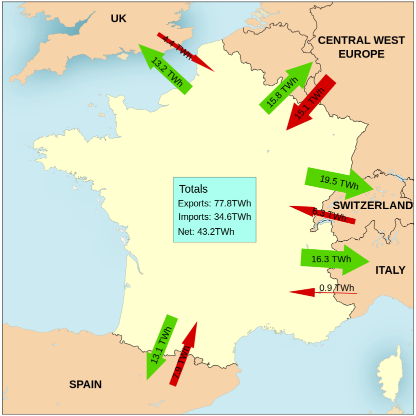
- As in most countries the electricity sector is competitive, generation is separated from transport and distribution.
- The transport (high voltage grid) is a natural monopoly and usually managed by Transportation Service Operator (TSO). In France the TSO is the RTE, who is responsible for managing the transport lines (above 50KV), real-time operation of the energy system and design of market mechanisms.
- The production is deregulated but the historical producer (EDF) has a market share of about 90% (private customers) and 65% (business customers).
- The distribution is carried out to 95% by ERDF, a regulated subsidiary of EDF.
- The production is mostly nuclear ($\approx 70\%$) but the share of renewables is increasing; large amounts of electricity are exported to other European countries.

The electricity sector in France

	Oil	Coal	Gas	Hydro	Nuclear	Wind	Solar	Biomass
2012	0.7%	3.6%	3.6%	13.7%	73.5%	2.9%	0.9%	1.3%
2022	0.5%	0.7%	9.9%	11.1%	62.6%	8.7%	4.2%	2.4%



Energy mix in France. Data source: RTE.

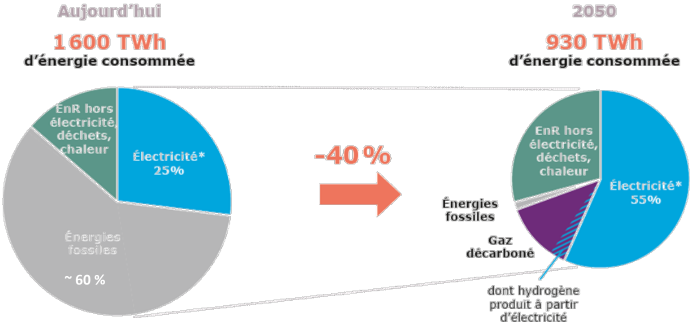


Export-import flows in France (2020).

Source: wikipedia (Template: Giro720; Data: RTE; Work: Auguel).

Future evolution of the French electricity sector: National Low Carbon Strategy (SNBC)

- First version adopted in 2015 mandated 75% reduction of CO2 emissions in 2050 (compared to 1990).
- New version adopted in 2020 defines a net zero objective in 2050.



Energy futures 2050: consumption scenarios

In octobre 2021, RTE published a large scale study “Energy futures”, defining alternative scenarios for French electricity consumption / production mix compatible with net zero in 2050

	HYPOTHÈSES	NIVEAU 2050	PRINCIPALES ÉVOLUTIONS
Référence	Électrification progressive (en substitution aux énergies fossiles) et ambition forte sur l’efficacité énergétique (hypothèse SNBC). Hypothèse de poursuite de la croissance économique (+1,3% à partir de 2030) et démographique (scénario fécondité basse de l’INSEE). La trajectoire de référence suppose un bon degré d’efficacité des politiques publiques et des plans (relance, hydrogène, industrie). L’industrie manufacturière croît et sa part dans le PIB cesse de se contracter. Prise en compte de la rénovation des bâtiments mais aussi de l’effet rebond associé.	**645 TWh**	180 TWh 134 TWh 113 TWh 99 TWh 50 TWh

	HYPOTHÈSES	NIVEAU 2050 (par rapport à la référence)	PRINCIPALES ÉVOLUTIONS (+ écart par rapport à la référence)
Sobriété	Les habitudes de vie évoluent dans le sens d’une plus grande sobriété des usages et des consommations (moins de déplacements individuels au profit des mobilités douces et des transports en commun, moindre consommation de biens manufacturés, économie du partage, baisse de la température de consigne de chauffage, recours à davantage de télétravail, sobriété numérique, etc.), occasionnant une diminution générale des besoins énergétiques, et donc également électriques.	**555 TWh** (-90 TWh)	160 TWh (-20 TWh) 111 TWh (-23 TWh) 95 TWh (-18 TWh) 77 TWh (-22 TWh) 47 TWh (-3 TWh)
Réindus-trialisation profonde	Sans revenir à son niveau du début des années 1990, la part de l’industrie manufacturière dans le PIB s’infléchit de manière forte pour atteindre 12-13% en 2050. Le scénario modélise un investissement dans les secteurs technologiques de pointe et stratégiques, ainsi que la prise en compte de relocalisations de productions fortement émettrices à l’étranger dans l’optique de réduire l’empreinte carbone de la consommation française.	**752 TWh** (+107 TWh)	239 TWh (+59 TWh) 134 TWh (0 TWh) 115 TWh (+2 TWh) 99 TWh (0 TWh) 87 TWh (+37 TWh)

Electricity consumption scenarios, source: RTE.

Energy futures 2050: generation mix scenarios

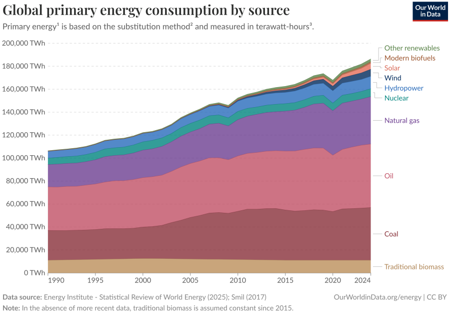
LES SCÉNARIOS DE MIX DE PRODUCTION
À L'HORIZON 2050

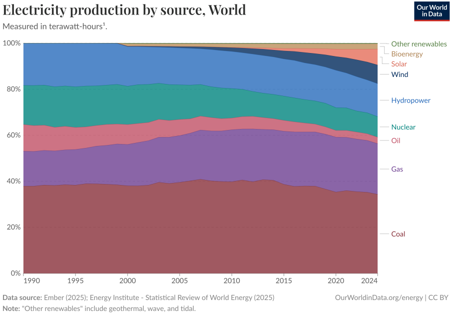
Filières : Flexibilités de la demande (hors V2G) · Véhicule-to-grid · Nouveau thermique décarboné · Batteries

	NARRATIF	RÉPARTITION DE LA PRODUCTION EN 2050	CAPACITÉS INSTALLÉES EN 2050 (EN GW)* Solaire	Éolien terrestre	Éolien en mer	Nucléaire historique	Nouveau nucléaire	BOUQUET DE FLEXIBILITÉS EN 2050
M0 100% EnR en 2050	Sortie du nucléaire en 2050 : le déclassement des réacteurs nucléaires existants est accéléré, tandis que les rythmes de développement du photovoltaïque, de l'éolien et des énergies marines sont poussés à leur maximum.	100% EnR; 9%, 21%, 31%, 36%	~208 GW (soit x21)	~74 GW (soit x4)	~62 GW	/	/	15 GW; 1,7 GW (1,1 MVE); 29 GW; 26 GW
M1 Répartition diffuse	Développement très important des énergies renouvelables réparties de manière diffuse sur le territoire national et en grande partie porté par la filière photovoltaïque. Cet essor sous-tend une mobilisation forte des acteurs locaux participatifs et des collectivités locales.	13% Nuc.; 87% EnR; 9%, 13%, 17%, 23%, 36%	~214 GW (soit x22)	~59 GW (soit x3,5)	~45 GW	16 GW	/	17 GW; 1,7 GW (1,1 MVE); 20 GW; 21 GW
M23 EnR grands parcs	Développement très important de toutes les filières renouvelables, porté notamment par l'installation de grands parcs éoliens sur terre et en mer. Logique d'optimisation économique et ciblage sur les technologies et les zones bénéficiant des meilleurs rendements et permettant des économies d'échelle.	13% Nuc.; 87% EnR; 9%, 13%, 21%, 32%, 22%	~125 GW (soit x12)	~72 GW (soit x4)	~60 GW	16 GW	/	15 GW; 1,7 GW (1,1 MVE); 20 GW; 13 GW
N1 EnR + nouveau nucléaire 1	Lancement d'un programme de construction de nouveaux réacteurs, développés par paire sur des sites existants tous les 5 ans à partir de 2035. Développement des énergies renouvelables à un rythme soutenu afin de compenser le déclassement des réacteurs de deuxième génération.	26% Nuc.; 74% EnR; 9%, 14%, 12%, 17%, 24%, 22%	~118 GW (soit x11)	~58 GW (soit x3,3)	~45 GW	16 GW	13 GW (soit 8 EPR)	15 GW; 1,7 GW (1,1 MVE); 11 GW; 9 GW
N2 EnR + nouveau nucléaire 2	Lancement d'un programme plus rapide de construction de nouveaux réacteurs (une paire tous les 3 ans) à partir de 2035 avec montée en charge progressive. Le développement des énergies renouvelables se poursuit mais moins rapidement que dans les scénarios N1 et M.	36% Nuc.; 63% EnR; 9%, 14%, 22%, 16%, 20%, 17%	~90 GW (soit x8,5)	~52 GW (soit x2,9)	~36 GW	16 GW	23 GW (soit 14 EPR)	15 GW; 1,7 GW (1,1 MVE); 5 GW; 2 GW
N03 EnR + nouveau nucléaire 3	Le mix de production repose à parts égales sur les énergies renouvelables et sur le nucléaire à l'horizon 2050. Cela implique d'exploiter le plus longtemps possible le parc nucléaire existant, et de développer de manière volontariste et diversifié le nouveau nucléaire (EPR 2 + SMR)	50% Nuc.; 50% EnR; 10%, 23%, 27%, 13%, 12%, 13%	~70 GW (soit x7)	~43 GW (soit x2,5)	~22 GW	24 GW	~27 GW (soit ~14 EPR + quelques SMR)	13 GW; 1,7 GW (1,1 MVE); 1 GW

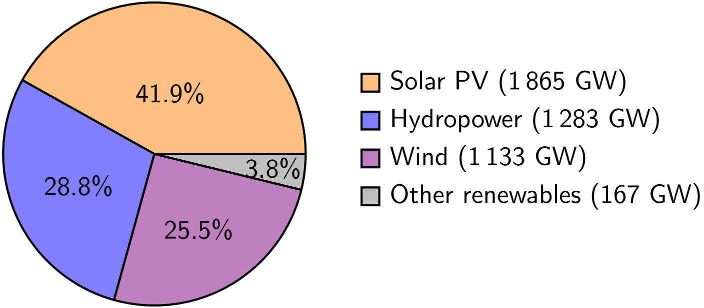
Hypothèses communes					
	Hydraulique ~22 GW	Énergies marines Entre 0 et 3 GW	Bioénergies ~2 GW	Imports 39 GW	STEP 8 GW

The energy/electricity sector in the world



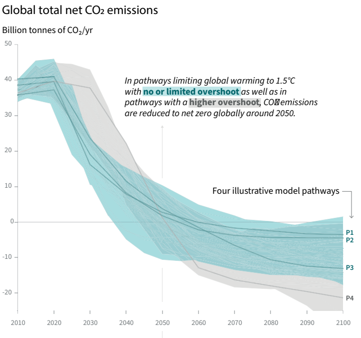


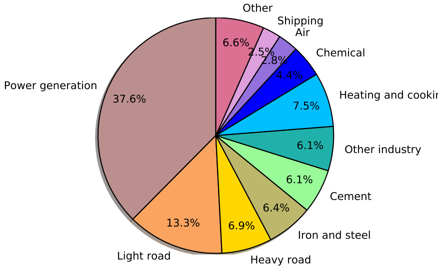
Renewable energy



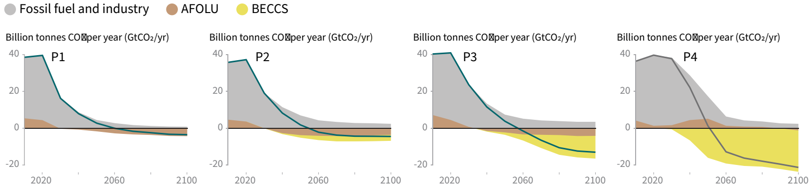
Global renewable power capacity by technology, end of 2024 (IRENA).

The energy transition





Share of emissions by source

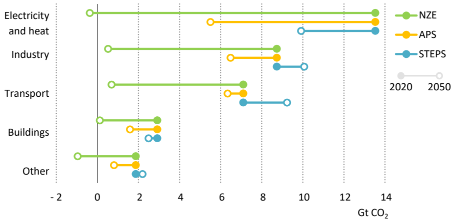


A pathway to net zero

- To stop climate warming below 1.5 degrees, carbon neutrality must be achieved in 2050-2060.
- Many countries have net zero engagements of various types and for various horizons.

Country	Year	Status
France	2050	Law
EU	2050	Political agreement
US	2050	Statement of intent
China	2060	Policy position

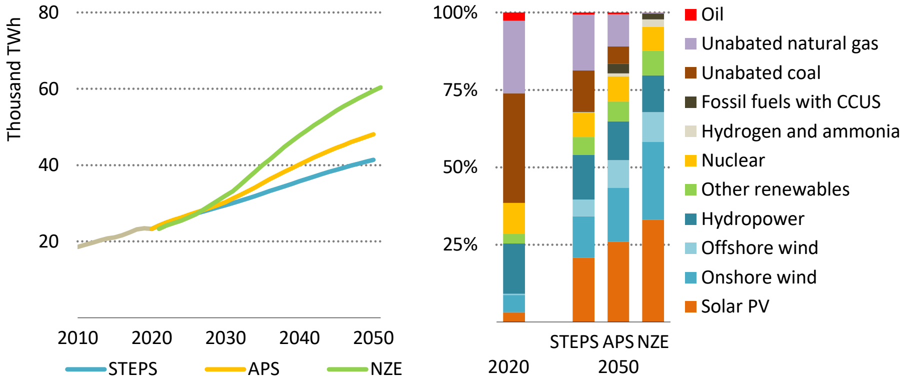
- International organizations are developing net zero scenarios for the energy sector, the most prominent being NZE 2050 scenario by IEA.



STEPS: Stated Policies Scenario ($\approx 2.6^{\circ}$ in 2100)
APS: Announced Pledges Scenario ($\approx 2.1^{\circ}$ in 2100)

Source: IEA World Energy Outlook 2021

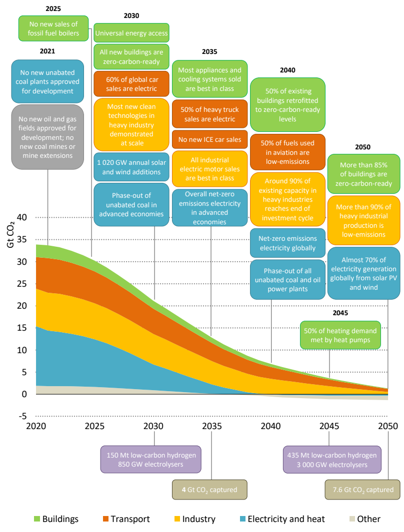
Global electricity demand and generation mix by scenario



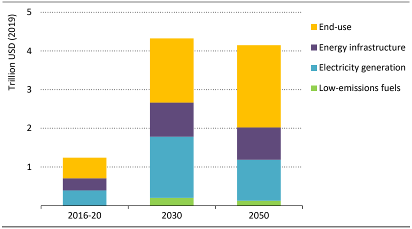
Source: IEA World Energy Outlook 2021

Setting near-term milestones and ramping up investment

Key milestones in the pathway to net zero



Clean energy investment in the net zero pathway



Source: IEA Net Zero by 2050

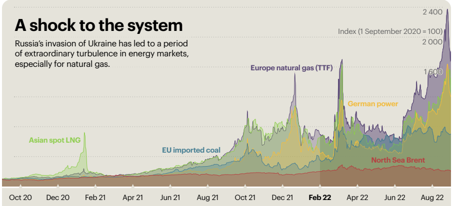
Russia-Ukraine war and the 2021-2022 energy crisis

- Unprecedented supply shock, exacerbated by growing energy demand due to post-pandemic recovery
- The rise of gas prices was the major cause of electricity price hikes since gas-fired plants are most often used as peak load generators
- Temporary increase in oil and coal-fired generation, increased CO2 emissions
- Long-term increase of investment into renewable energy (REPowerEU and Inflation Reduction Act = over USD 1.2 trn), energy efficiency and economy measures and in some cases LNG facilities

$\Rightarrow$ Increased energy security and reduced CO2 emissions in the long term

Attention to locked-in emissions.

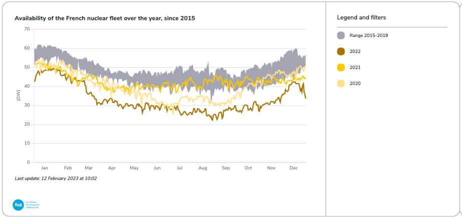
Evolution of energy prices in 2020-2022, source: IEA $\rightarrow$

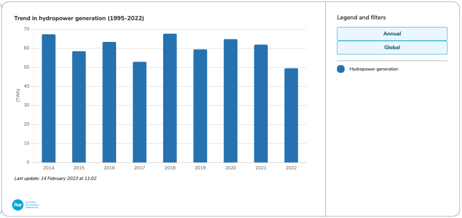


The 2021-2022 energy crisis in France

Three independent but simultaneous crises which compounded one another:

- Soaring gas prices in the wake of Russia's invasion of Ukraine.
- A crisis of nuclear power after discovery of a generic fault affecting recent reactors. Output was 30% below the average of past 20 years;
- A drought drove hydropower output down to its lowest level since 1976.

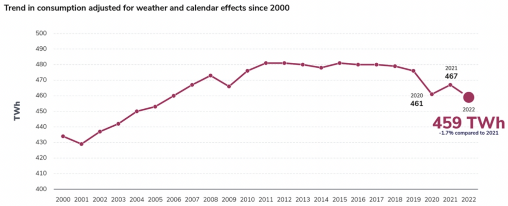




The 2021-2022 energy crisis in France

The effects of the crisis were mostly economic:

- There was no supply disruption due to lower demand, but France became a net importer of electricity (4% of consumption) for the first time since 1980
- Consumption was down 4% compared to 2014-2019 average due to reduced demand from the industrial sector and energy saving measures
- The year saw record additions of renewable capacity (1.0 GW of onshore wind; 2.6 GW of solar; first offshore wind power plant of 0.5 GW)



Aftermath of the crisis

- In the short term the natural was partly replaced by coal, leading to 4% increase in CO2 emissions from European power sector
- In the medium term, many new deals were sighed to diversify oil and gas supplies and make them secure, and EU energy platform aiming to limit prices through joint gas purchases was established
- In the long term, initiatives were taken at EU and national level to accelerate the energy transition which may have been brought forward by 10 years
- However, the development of renewable energy is slowed down by high interest rates, price caps and legal constraints

The Economist

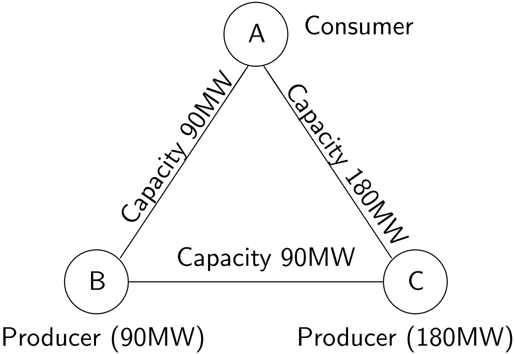
Outline

Features of electricity as a commodity

- Electricity is a local commodity: cross-border capacity is limited, market structure is different in different states
- Electricity network is international: it is subdivided into multiple energy systems (frequency control zones), encompassing multiple countries
- Electricity is almost non-storable yet demand must equal supply at all times within each frequency control zone
- To ensure sufficient supply, reserves must be put aside at different time horizons and adequate market structure must be created
- Electricity infrastructure required heavy long-term investment, not all of it can be finances through traditional energy markets

Electricity transport networks

- Sum of currents flowing in and out of each node is zero
- Sum of voltages around each loop is zero
- Electricity flows through all available paths



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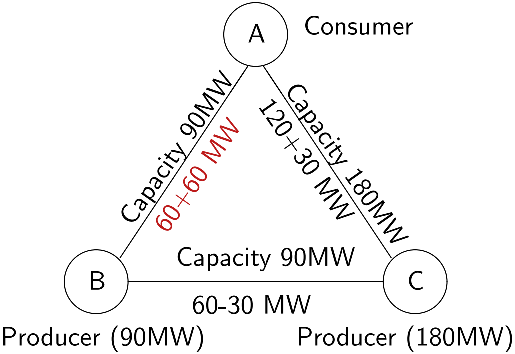


Reactive Power

- In AC, voltage and current are sinusoids with phase shift φ:

$$v(t) = \hat{V} \cos \omega t, \quad i(t) = \hat{I} \cos(\omega t - \varphi)$$

- Active power (useful work, net energy per cycle, measured in watts):

$$P = \frac{1}{T} \int_0^T v(t) i(t) dt = \frac{1}{2} \hat{V} \hat{I} \cos \varphi = VI \cos \varphi,$$

  where $V = \frac{\hat{V}}{\sqrt{2}}$ and $I = \frac{\hat{I}}{\sqrt{2}}$ are root mean square voltage and current.

- Reactive power (measured in volt-amperes)

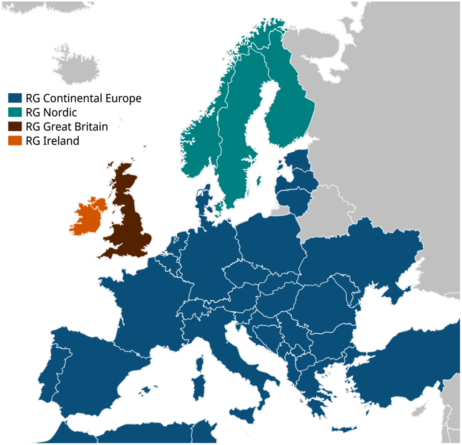
$$Q = \frac{1}{T} \int_0^T v(t) i_\perp(t) dt = \frac{1}{2} \hat{V} \hat{I} \sin \varphi = VI \sin \varphi, \quad i_\perp(t) = \hat{I} \sin(\omega t - \varphi)$$

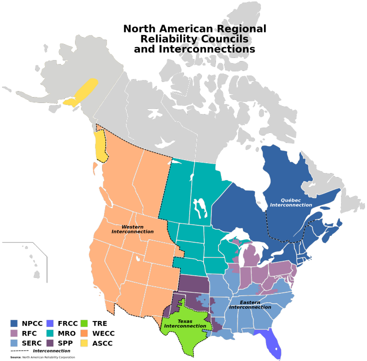
  $\Rightarrow$ Energy is *stored* in magnetic/electric fields (inductors, capacitors) during part of the cycle, then *returned* to the grid in the next part: no net energy transfer over one period.

Reactive Power

- Inductive loads ($\varphi > 0$): $Q > 0$ (consume reactive power).
  Capacitive loads ($\varphi < 0$): $Q < 0$ (supply reactive power).
- Apparent power $S = \sqrt{P^2 + Q^2}$ uses current and line capacity
  $\Rightarrow$ reactive power affects flows, losses and voltage limits, even though it does not produce useful work.
- Transport lines mostly consume reactive power
- Large synchronous generators can adjust their power factor ($\cos\varphi$).
- Reactive power generators (capacitor banks) are used at distribution level.
- Reactive power is currentlty procured as a locational ancillary service but more explicit market-oriented designs are being developed and tested.

Frequency control zones

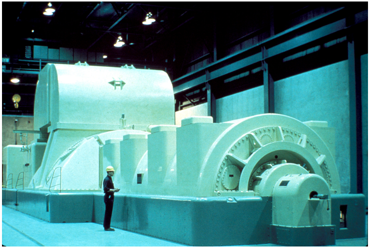




Continental Europe (plus Turkey, Morocco, Algeria and Tunisia) is a single frequency control zone. Ireland, UK and Texas are AC islands.

Frequency control

- Network stability over very time horizons (seconds) is ensured by frequency control.



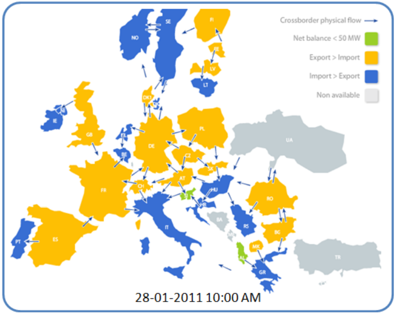
- Due to inertia of conventional generators, after a sudden load increase (power plant failure), frequecy of the AC system goes down progressively, allowing the automatic frequency control systems to ramp up production.

- Renewable generators have no spinning reserve: increased renewable penetration makes frequency control more difficult.

- The effect is stronger in "AC islands" with strong renewable penentration: a study (Connolly et al., 2010) has shown that the energy system of Ireland cannot operate safely with wind energy penetration of over 50% and that the optimal penetration is between 21% and 36%.

European interconnections

- Primary purpose of interconnections: ensure system reliability by pooling frequency reserve and providing emergency assistance in case of outage of generation / trasmission facility.
- The interconnections also enable power suppliers to sell their energy to customers in another country of the European Union.



Cross-border physical flows in Europe. Source: `entsoe.net`

- Actual transmission capacities are different from physical capacities and are determined by the system operators based on generation and load patterns.

European interconnections

- Available capacity is allocated on annual, monthly, daily and intraday basis.
- Market coupling mechanism (ensuring the same price in different markets) involves implicit capacity allocation.
- Users first acquire capacity rights (allocation step) then declare to TSOs the program they wish to implement within the capacity acquired (nomination step).
- Since 2007, transmission rights may also be bought in the secondary market.

Outline

Distribution and network-related risks

Contingencies

- Load fluctuations
- Meteorological hazards (lightning, storms, frost, floods, cold, etc.)
- Outages and external hazards (accidents damaging electrical installation)
- Human errors in operation and maintenance

Vulnerabilities

- Cascading failures of transmission lines
- Voltage collapse
- Frequency collapse
- Loss of network synchronicity

Stages of network collapse: cascading failures

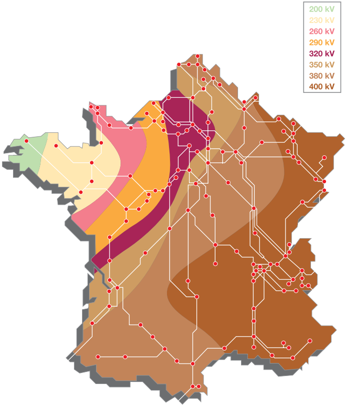
- Overloaded lines will disconnect, increasing the load on other installations, which will in turn disconnect themselves, provoking a cascading failure.
- Operators estimate loads in case of failure of one component ($N - 1$ security rule) and prepare measures to restore $N - 1$ security in case of failure.

Cascade tripping of Italy-Europe UHV lines on 28 Sep 2003.

Source: UCTE report on the 28 September 2003 blackout in Italy

Stages of network collapse: voltage collapse

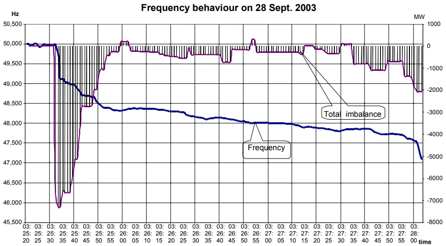
- Voltage is a local characteristic of the network, controlled by load-sensitive transformers, capacitors and local generators.
- In case of load increase, load-sensitive transformers automatically modify their transformation ratios, to maintant constant customer voltage.
- This increases the load and lowers the voltage on the transport network, reducing the transmissible power, and leading to voltage collapse if no measures are taken.



Isovoltage curves on the 400 kV grid during the incident of 12 January 1987. Source: RTE

Stages of network collapse: frequency collapse

- Frequency is a global charactereistic of power networks
- Frequency stability expressed the balance between generation and load: if load exceeds generation, the speed of turbines starts to drop and the frequency goes down
- In France, admissible frequency range is 50 Hz ± 0.5 Hz. Automatic load shedding starts at 49 Hz.

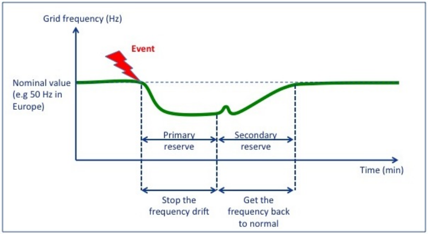


Frequency collapse during the 2003 blackout in Italy.

Source: UCTE report on the 28 September 2003 blackout in Italy

Defense against frequency collapse

- Primary control: automatic turbine valve controllers restore supply demand balance at a lower frequency level by increasing output;
- Secondary control: capacity of generator units is increased (by the national dispatch center) to bring frequency back to normal.

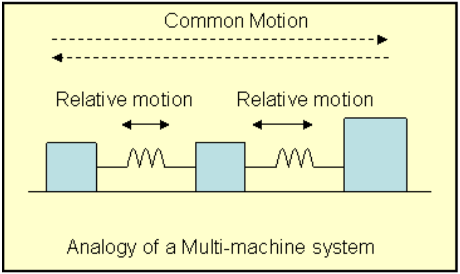


Source: `cleanhorizon.com`

- Tertiary control: balancing bids are activated to reconstitute primary / secondary reserve.
- In France automatic load shedding takes place at thresholds of 49, 48.5, 48 and 47.5 Hz (20% at each level).

Stages of network collapse: loss of synchronicity

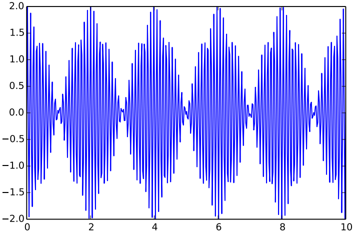
- In a stable system, all generators run at the same speed.
- This happens because interconnected synchronous generators produce restoring torques when subjected to small disturbances from the equilibrium.
- Existence of large synchronous regions allows to pool primary frequency reserve and provide emergency assistance between neighbors.



Source: `nptel.ac.in`

Stages of network collapse: loss of synchronicity

- Under exceptionally strong disturbances (short circuit etc.), due to nonlinear effects, the coupling between the generators may be broken and synchronicity may be lost, resulting in voltage and current spikes.
- Due to action of protection systems, the facilities should then 'gracefully' separate forming autonomous islands.
- Sometimes this does not happens fast enough and the facilities go offline before being separated, leading to partial or full blackout.



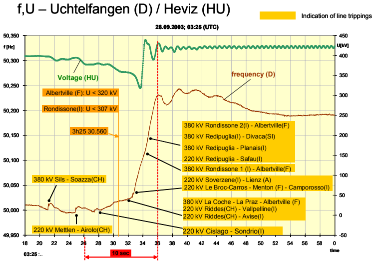
Voltage on a constant resistance load connected to two parallel generators rotating at speeds of 50Hz and 50.5Hz.

Illustration: Switzerland-Italy incident of 28 Sep 2003

3AM: Exchange capacity of Italy with Switzerland and rest of Europe is saturated

3:01:42: Tripping of the Mettlen-Lavorgo 380 kV line (running at 86% capacity). The Sils-Soazza line is overloaded to 110%; the operator has 15 minutes to decrease load.

3:11: Swiss operator asks the Italian one to reduce the load by 300MW, which is executed 10 minutes later. This is sufficient to eliminate the overload under favorable conditions but does not restore $N-1$ security



Voltage and frequency recorded in Germany and Hungary during the starting phase of the blackout.

Source: UCTE report on the 28 September 2003 blackout in Italy

3:25:21: Tripping of the Sils-Soazza (SW) line followed (in 4 seconds) by the Airolo-Mettlen (SW) line and the Lienz (AU)-Soverzene line

3:25:28: Loss of synchronism and separation of Italy from the rest of Europe

Illustration: Switzerland-Italy incident of 28 Sep 2003

The disconnection of Italy from the UCTE system causes power and voltage swings; frequency falls to 49.1 Hz.

Due mainly to voltage collapse, 21 of 50 main generation units disconnect: islanding of Italy is unsuccessful

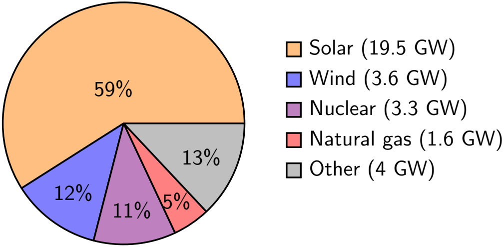
2mn30s after the separation the Italy network is completely dead.

4:05 AM to 12:45 PM: gradual reconnection of Italy with neighbors

6AM to 4PM: restoration of supply

Spain and Portugal blackout of April 28, 2025

- On Monday, 28 April 2025, at 12:33 CEST, a major power blackout originating in Spain affected Spain and Portugal.
- Power was not fully restored until April 29, at 00:22 in Portugal and 04:00 in Spain.



Power generation in Spain before the blackout

- Before the blackout, the power generation in Spain was predominantly wind and solar, and Spain was exporting energy to Portugal, France and Morocco.
- Details of the incident are descibed in ICS Investigation Expert Panel report (October 3rd) but causes are not yet fully known

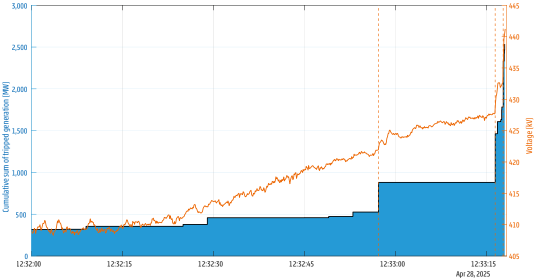
Sequence of events

- Between 12:32:00 and 12:32:57, there was a loss of 208 MW distributed wind and solar generators in northern and southern Spain, as well as an increase in net load in the distribution grids of approximately 317 MW, which might be due to the disconnection of small embedded generators or to an actual increase in load or to a combination of both. The reasons for these events are not known.
- From 12:32:57 until 12:33:18, major disconnection events occurred in the regions of Granada, Badajoz, Sevilla, Segovia, Huelva, and Cáceres, which resulted in an additional loss of generation of at least 2 GW. Some of these trips occurred due to overvoltage protection, but the reasons of most of trips are not known.
- As some generation units were consuming reactive power with the effect of reducing the voltage, the disconnections of these units without adequate compensation of loss of reactive power by other resources in the system with the capability to inject/absorb reactive power meant that voltages in the system increased, not only in Spain but also in Portugal. Furthermore, the frequency decreased.

Sequence of events

- Between 12:33:18 and 12:33:21, the voltage of the South area of Spain sharply increased, and consequently also in Portugal. The over-voltage triggered a cascade of generation losses that caused the frequency of the Spanish and Portuguese power system to decline.

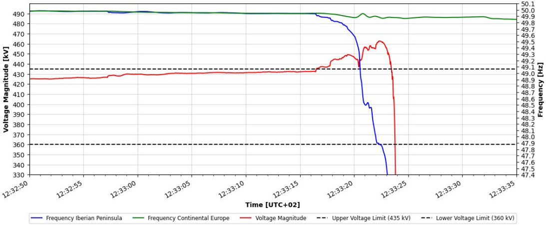
- At 12:33:19, the power systems of Spain and Portugal started losing synchronism with the rest of the European System.



Cumulative sum of tripped generation vs voltage in Spain

Sequence of events

- Between 12:33:19 and 12:33:22, the automatic load shedding and System Defence Plans of Spain and Portugal were activated but unable to prevent the collapse of the Iberian power system.
- At 12:33:20.473, the AC interconnection to Morocco tripped due to underfrequency.
- At 12:33:21.535, the AC overhead lines between France and Spain were disconnected by protection devices against a loss of synchronism.
- Finally, at 12:33:23.960 all system parameters of the Spanish and Portuguese electricity systems collapsed.



Frequency and voltage in Spain and of frequency in the rest of Europe

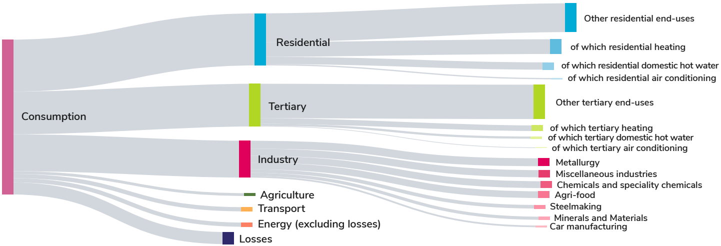
Restoration process

- Until 15:30, eight black-start islands could be built up successfully in Spain and first areas were already connected to rest of CE via interconnectors between Spain and France
- At 13:04, the interconnection between Spain and Morocco was energised, which had to be repeated at 14:34 after tripping at 14:27.
- At 18:36, the interconnector between Spain and Portugal was energised and the Portuguese transmission grid received voltage with continental frequency again. In the meantime, two black-start islands could be built up in Portugal.
- At 19:32, the Southern zone connected with Moroccan system could be synchronised with the Northern zone connected to CE.
- At 00:22 and around 04:00 on 29 April 2025, the restoration process of the transmission grid was completed in Portugal and Spain, respectively.

Outline

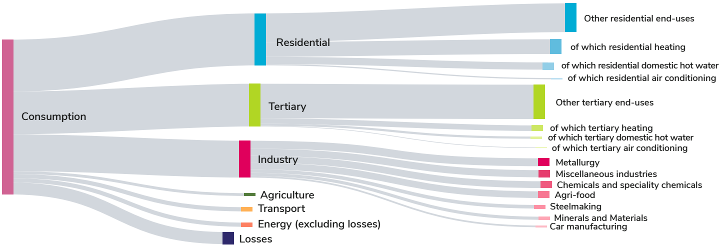
Electricity consumption in France

Structure of electrical consumption in France (459.3 TWh in 2022)



Electricity consumption in France

Structure of electrical consumption in France (459.3 TWh in 2022)

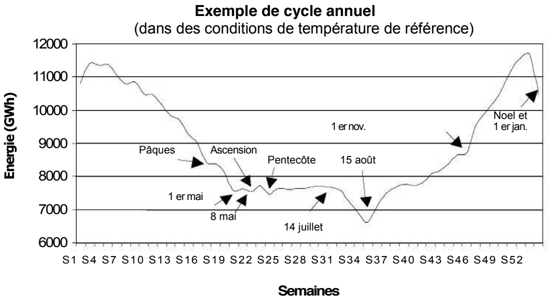


Features of electrical consumption:

- Strongly seasonal with pronounced yearly, weekly and seasonal cycle
- Strongly correlated with meteorological conditions
- Non-stationary with trends depending on the evolutions of electricity usage
- Relatively easy to forecast 1-2 days ahead

Electricity demand: seasonal trends

- In France, maximal consumption is observed in December and minimal around August 15

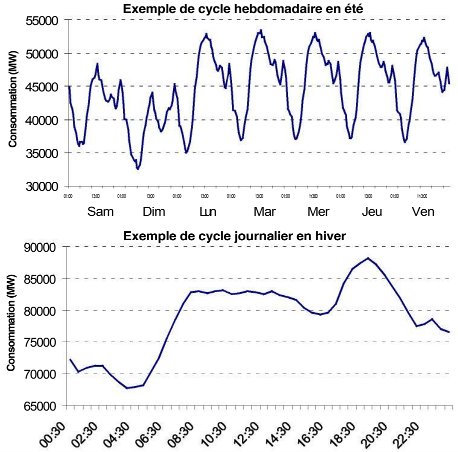


Source: RTE, Consommation française d'électricité, Nov 2014

- The consumption is lower during the week-end and especially on Sunday due to reduced economic activity
- The daily maximum is attained around 1PM in summer and 7PM in winter; the minimal consumption is observed during the night.

Electricity demand: meteorological effects

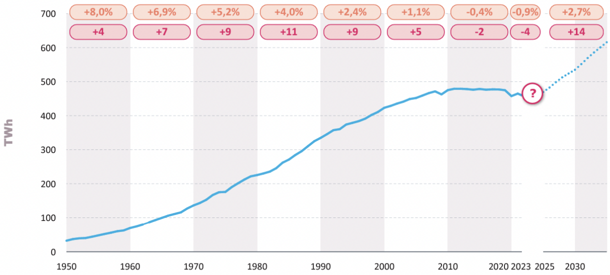
- Temperature: in France, in winter, a variation of $1^\circ C$ corresponds to an variation of consumption of 2500 MW. In summer, the variation is about 400 MW due to air conditioning.
- Cloud cover: measured on the scale from 0 to 8. A variation of 1 unit corresponds to a variation of consumption of about 800 MW.



Source: RTE, Consommation française d'électricité, Nov 2014

Electricity demand: non stationarity

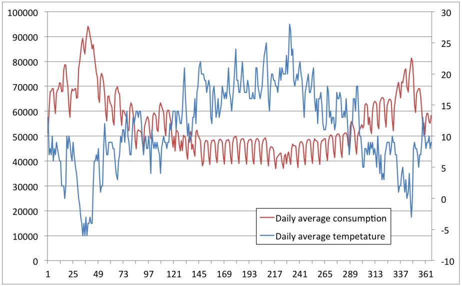
- Electrical consumption is not stationary: average annual consumption has been rising throughout the 20th century, but has stabilized recently in France and has even fallen slightly in the EU over the last 10 years.
- It is expected to rise again due to electricication of carbon-intensive activities.



Electrical consumption in continental France corrected for weather and calendar effects. Source: RTE, Bilan previsionnel 2023

Electricity demand: consumption spikes

- The main risk for power systems comes from *consumption spikes* during the cold season, which are caused by cold waves.



Data source: RTE web site and `www.wunderground.com` (temperature at Paris Orly)

Electricity demand: risks for TSO

- The risks of electricity consumption are related to ensuring supply-demand balance and preventing blackouts.
- In the long term, network structure must be adapted to new consumption patterns, new plants and interconnections must be built.
- In the medium term, sufficient supply margin must be ensured for the cold season, taking into account the possibility of extreme tempetatures.
- In the short term (1-2 days), sufficient production units must be affected to meet demand.
- Following the liberalization of electricity markets, the supply-demand balance is managed through market mechanisms in a (partially) decentralized manner.
- In France, specific tools have been developed by RTE: perimeter of balance, capacity mechanism, adjustment mechanism.

Managing supply-demand balance

- Balance responsible entity is a large consumer / producer ($\approx$ 150 in France) responsible for ensuring supply-demand balance within his balance perimeter. He compensates RTE for negative imbalances within his perimeter and receives compensation for positive imbalances. He has access to the block exchange service of RTE allowing to trade blocks of electricity with other responsible entities.
- Capacity mechanism: obligation on suppliers to ensure spare capacity corresponding to peak consumption of their clients. Extra capacity is compensated by RTE.
- Adjustment mecanism: a market for very short term (15 minute) offers designed to ensure the generation / demand balance in case of unexpected incidents.

Load management

- Individual customers may be encouraged to disconnect loads via specific tariffs (EJP - effacement jour de pointe in France).
- Industrial customers may post load shedding offers on the adjustment markets.
- Utility companies may disconnect certain loads (heating, air conditioning, electric car chargers) directly via smart meters.

Demand response

- Contract between a consumer and a producer (or a retailer)
- The consumer pays a lower fare for power all the days of the year except on a certain days (or periods) decided by the retailer.
- The number of days of price events is determined at inception.
- This form of demand-response contract is named dynamic Time-Of-Use (dToU).

Example: Low Carbon London Pricing trial experiment 2012-2013

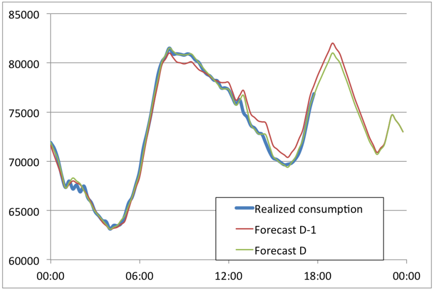
- 5,567 London households with consumption measured at an half hourly time-step on the period from February, 2012 to February, 2014.
- Standard tariff was 14 p/kWh.
- Consumer enrolled in the dToU tariff would pay their power:
  - 11.76 p/kWh on Normal days,
  - 67.2 p/kWh on High price days,
  - 3.99 p/kWh on Low price days.

Demand dispatch

- Demand dispatch refers to direct control of loads by the system operator to optimize network operations.
- The controllable load must be able to shift their consumption in time without affecting the end users. Examples are : water heaters, heating, ventilation, and air conditioning (HVAC), electric vehicle chargers, pool pumps etc.
- Mechanisms for remotely controlling the loads and incentivizing the consumers are presently being developed.

Load forecasting and scenario generation

- Short-term forecasting demand (up to 1 week) is based on meteorological forecasts and on historical consumption data.
- Seasonal forecasts are produced with reference meteorological data. Historical simulations may be used for scenario analysis.



Data source: RTE web site

- Long-term consumption scenarios produced by RTE describe how electrical consumption should evolve to meet energy transition objectives (cf Energy Futures exercise).